

inputs

attributes x_{ij}

covariates z_i

history h_{ij}

1 structural utility

$$V_{ij} = \text{ASC}_j + \beta^\top x_{ij} \\ - \beta_t(z_i)t_{ij} - \beta_c(z_i)c_{ij} + g_j(z_i) + u_{p(i)j} \\ R = 5 \text{ fits averaged in probability}$$

τL_{ij}

2 boosted residual

$$U_{ij} = \tau L_{ij} + f_j(x_{ij}, z_i, h_{ij}) \\ \text{one ensemble per alternative,} \\ \text{monotone in time and cost}$$

$1 - \pi$

$$q_i(j) = \text{softmax}_j(a U_i)$$

3 language channel

frozen
LLM

K sentences
per alternative

frozen encoder
768-d

reader
 $\rightarrow \bar{p}_i(j)$

no gradient reaches the language model or the encoder

probability mixture

$$p_i(j) = (1 - \pi) q_i(j) \\ + \pi \bar{p}_i(j)$$

π and a stacked
on out-of-fold predictions